01 - CYBERSECURITY AND CIA TRIAD

1. Cybersecurity?

Cybersecurity is the practice of protecting computers, servers, networks, applications and digital information from unauthorized access, cyberattacks, damage and theft.

In simple words, cybersecurity means keeping our digital devices and information safe.

Example: Using a strong password and two-factor authentication to protect an email account.

Objectives of Cybersecurity

- Protect sensitive and personal information.
- Prevent unauthorized access to computer systems.
- Detect cyberattacks and suspicious activities.
- Protect networks, applications and devices.
- Reduce security risks.
- Recover systems and data after security incidents.

Common Cybersecurity Threats

Threat	Meaning	Example
Phishing	Tricking people into sharing sensitive information	A fake bank email asks for a password
Malware	Software designed to cause harm	A malicious program infects a computer
Ransomware	Malware that encrypts files and demands payment	A company loses access to its documents
Password attack	Attempting to obtain or guess passwords	An attacker guesses a weak password
Data breach	Unauthorized access to protected information	Customer records are stolen
Denial-of-Service (DoS)	Making a service unavailable by overwhelming it	A website cannot respond to legitimate users

2. CIA Triad?

The CIA Triad is a fundamental information security model consisting of three principles:

- **C — Confidentiality**
- **I — Integrity**
- **A — Availability**

It helps organizations protect data and keep computer systems secure and reliable.

3. Confidentiality

Confidentiality means ensuring that information is accessible only to authorized people, systems or processes.

In simple words, confidentiality means **keeping information private**.

Real-world example

A hospital stores patient medical records. Only authorized doctors and staff should be able to access those records.

Methods to maintain confidentiality

- Use strong passwords.
- Enable multi-factor authentication (MFA).
- Apply access permissions.
- Encrypt sensitive data.
- Avoid sharing confidential information with unauthorized people.

What happens when confidentiality fails?

An unauthorized person may view or steal private information, such as passwords, bank details or medical records.

Example of failure: An attacker gains access to an employee's account and reads confidential company documents.

4. Integrity

Integrity means maintaining the accuracy and completeness of information and protecting it against unauthorized modification or deletion.

In simple words, integrity means **keeping information correct and unchanged unless an authorized change is made**.

Real-world example

A bank's transaction records must accurately show the money deposited and withdrawn from a customer's account.

Methods to maintain integrity

- Apply appropriate file permissions.
- Use access controls.
- Maintain audit logs.
- Use cryptographic hashes to detect changes to files.
- Keep secure backups and verify data.

What happens when integrity fails?

Information may be altered, deleted or manipulated without authorization.

Example of failure: An unauthorized person changes a student's examination marks in a database.

5. Availability

Definition

Availability means ensuring that information, applications and systems are accessible to authorized users when required.

In simple words, availability means **making sure systems are working when people need them**.

Real-world example

An online banking application should be available when customers need to transfer money or check their account balances.

Methods to maintain availability

- Maintain regular backups.
- Install security updates.
- Monitor system performance.
- Use redundant systems where appropriate.
- Protect services against denial-of-service attacks.
- Prepare disaster recovery plans.

What happens when availability fails?

Authorized users cannot access a system or information when they need it.

Example of failure: A hospital's computer system becomes unavailable because of a server failure, delaying access to patient records.

6. CIA Triad Comparison Table

Principle	Main Goal	Real-World Example	Security Failure
Confidentiality	Keep information private	Protect medical records	Unauthorized disclosure
Integrity	Keep information accurate	Protect bank transaction records	Unauthorized modification
Availability	Keep systems accessible	Maintain hospital systems	Service outage

Easy trick to remember

- **Confidentiality:** Who can see the information?
- **Integrity:** Is the information correct?
- **Availability:** Can authorized users access it?

Memory trick: Private — Correct — Accessible.

7. CIA Triad in Online Banking

Consider a mobile banking application.

Confidentiality: Only authorized users can access protected account information.

Integrity: Account balances and transaction records must remain accurate.

Availability: Customers should be able to access the banking service when required.

If any one of these principles fails, the security or reliability of the banking service is affected.

8. Difference Between Cybersecurity and the CIA Triad

Cybersecurity	**CIA Triad**
A broad field focused on protecting digital systems and information	A basic model used to explain three important security objectives
Includes technologies, processes, policies and people	Includes Confidentiality, Integrity and Availability
Covers prevention, detection, response and recovery	Helps identify what security must protect
Example: Installing security updates and monitoring a network	Example: Ensuring data is private, correct and accessible